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PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
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Flight Program:			
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COI Name (Institution):	Demetriou, Marios (California Institute of Technology) Johnson, William (California Institute of Technology) Rhim, Won-Kyu (California Institute of Technology) Shapiro, Andrew (Jet Propulsion Laboratory)		
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Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. The investigators propose to study the nucleation, growth, directionality and cooling rate dependence of crystalline dendrites evolving in a glass-forming eutectic matrix. The ground-based research effort will be carried out utilizing laboratories at the Jet Propulsion Laboratory and the California Institute of Technology, including the High Vacuum ElectroStatic Levitator platform (HVESL) at Caltech. The investigators propose a four-component ground-based project consisting of (1) directional solidification experiments utilizing Moving Mushy Zone (MMZ) and Bridgman solidification, (2) morphological characterization of dendrite growth and orientation during "mushy" (or "semi-solid") processing, (3) mechanical testing and development of structure-property relations, and (4) dendrite formation and growth during containerless processing utilizing the HVESL. After completion of a ground-based research program, the investigators will extend the experimentation to flight-based microgravity research aboard the ISS, the aim of which is to study gravity-driven convection arising from density gradients in the microstructure. The flight experiment objectives are to (1) use the Low Gradient Furnace (LGF) to complete Bridgman crystal growth on composites in an attempt to grow dendrites along the translational axis, (2) use the Solidification and Quenching Furnace to coarsen and homogenize dendrite microstructures using mushy-zone processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
Task Progress:	The objectives of the research proposal that were completed during FY12 were fourfold: 1. Processing Equipment Development 2. Morphology of Dendritic Microstructures as a Function of Processing 3. Correlating Mechanical Properties with Microstructures and Processing 4. Mushy Processing using Electrostatic Levitation Platform To complete Objective 1, we successfully finished the new JPL Metallurgy Facility, which is capable of developing and studying new metal alloys and composites. We also developed a vacuum furnace to replicate conditions on the Materials Science Research Rack on the ISS to do directional solidification. To determine viscosity of composites samples, we fabricated wire drawing and parallel plate viscometry setups. To complete Objective 2, we studied a new CuZrAlCo alloy and the role of cooling on microstructure. We also performed infiltration experiments on composites samples with varying density of inclusions. To complete Objective 3, we performed cryogenic Charpy impact testing on a variety of composite samples to study the effect of temperature and processing on toughness. To complete Objective 4, we did isothermal holds to coarsen microstructures using the ESL system and we looked a high temperature viscosity of the same composites in the liquid regime. An article entitled “Consequences of polymorphic crystallization in a transformation induced plasticity bulk metallic glass matrix composite” is under preparation to be submitted to the Journal of Materials Research.

Bibliography Type:	Description: (Last Updated: 01/01/2026)
Abstracts for Journals and Proceedings	Hofmann DC. "Microgravity Research on Bulk Metallic Glasses and Composites." TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. Conference program, p. 183. http://www.tms.org/Meetings/Annual-12/PDFs/TMS2012FinalProgramWEBwithAbstracts.pdf , Mar-2012
Articles in Peer-reviewed Journals	Roberts S, Zachrisson C, Kozachkov H, Ullah A, Shapiro A, Johnson W, Hofmann DC. "Cryogenic Charpy impact testing of metallic glass matrix composites." Scripta Materialia 2012 Mar;66(5):284-7. http://dx.doi.org/10.1016/j.scriptamat.2011.11.011 , Mar-2012
Articles in Peer-reviewed Journals	Kozachkoz H, Kolodziejska J, Roberts S, Li J, Johnson W. "Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices using Electrostatic Levitation." Scripta Materialia, under review, July 2012. Expected publication September 2012. , Jul-2012